DBSCAN Clustering and Outlier Detection

Exploring density-based clustering with DBSCAN on concentric-ring data and the Iris dataset to demonstrate both shape-flexible clustering and outlier detection.

Jibril Hussaini | Python · scikit-learn · NumPy · matplotlib

Overview

This project investigates DBSCAN (Density-Based Spatial Clustering of Applications with Noise), a clustering algorithm that groups points by local density rather than distance to a centroid. Two datasets are used: a synthetic two-circles dataset that defeats k-means, and the classic Iris dataset augmented with 20 uniformly injected outliers. The analysis covers parameter selection, cluster recovery, and the use of the noise label as an outlier-detection mechanism.

Goals

Demonstrate where and why k-means fails on non-convex geometry, show how to choose the DBSCAN epsilon parameter objectively, and evaluate DBSCAN as a practical outlier detector under increasing contamination.

Objectives

- Generate a two-circles dataset and show that k-means cannot recover the concentric rings while DBSCAN can.
- Select epsilon objectively from the knee of the sorted k-distance graph using Schubert's heuristic.
- Apply DBSCAN to a 4-dimensional Iris dataset with 20 injected uniform outliers and compare cluster assignments to k-means.
- Measure outlier-detection recall and precision and track how they degrade as the contamination fraction increases.
- Interpret the DBSCAN noise label (-1) as an unsupervised outlier flag and quantify its reliability.

Approach

For the circles experiment, a NearestNeighbors model is fitted with k=4 to compute each point's 4th-nearest-neighbour distance; sorting these in descending order reveals a clear knee that guides epsilon selection. On Iris, 20 uniform random points are injected outside the feature ranges and DBSCAN is run with an analogously chosen epsilon. K-means is run in parallel as a baseline to highlight its inability to ignore noise points. Outlier recovery is measured by treating the injected points as the positive class.

Outcomes

- DBSCAN with eps=0.18 achieved a high adjusted Rand index on the two-circles data, successfully separating both rings where k-means scored near zero.
- On Iris, DBSCAN flagged almost all 20 injected outliers as noise (high recall and precision) while k-means assigned every outlier to one of its three forced clusters.
- Outlier-detection recall remained high up to roughly 20-40 percent contamination; beyond that the

injected points formed artificial density pockets and began to be absorbed as clusters.

- The k-distance knee provided a reproducible, data-driven basis for epsilon selection without any labelled data.